

MANGENESIS

AUTONOMOUS MINERAL INTELLIGENCE & 4D DIGITAL TWIN

Comprehensive Architectural Re-Engineering Blueprint specifying the next-generation distributed platform for multi-mine operations, spaceborne InSAR and multispectral analytics, volumetric ore body modeling, real-time IoT fleet telematics, and predictive production continuity.

CORE ARCHITECTURAL PILLARS

1. WebGPU & 4D Spatial Visualization Engine

Hardware-accelerated subterranean voxel rendering, PBR strata shaders, and 60 FPS multi-mine digital twins.

2. Real-Time High-Throughput Ingestion (Go / Rust)

50k+ events/sec streaming CAN-bus fleet telemetry, seismographs, and ISRO/NASA satellite passes.

3. Spatial & Time-Series Data Lakehouse

PostgreSQL 17 + PostGIS 3.5, TimescaleDB, and Apache Sedona for distributed multi-spectral raster queries.

4. AI Volumetric Grade Modeling & Kriging

Sparse 3D-CNNs + Ordinary Kriging for UNFC 111 drill assay interpolation & LightGBM production forecast.

DOCUMENT METADATA

Client: MOIL Limited (Min. of Steel, Govt of India)

Target Platform: Multi-Mine Autonomous Fleet & Reserve System

Architecture Version: 2.0-Enterprise (Q3 2026)

Classification: Highly Confidential / Technical Specification

1. Executive Summary & Problem Diagnosis

The existing mining operations at MOIL open-pit and underground locations (Gumgaon, Dongri Buzurg, Mansar, Ukwa, and Balaghat) rely on fragmented legacy systems: 2D desktop GIS applications, offline spreadsheets, infrequent manual survey cross-sections, and reactive equipment maintenance. This operational fragmentation leads to 12-18% preventable production shortfalls, suboptimal grade blending, and delayed slope instability detection.

MANGENESIS 2.0 re-engineers the operational lifecycle into a unified, event-driven, 4D Geospatial Intelligence Platform that bridges orbital remote sensing, subsurface geostatistics, and autonomous haulage dispatch.

ARCHITECTURAL PARADIGM SHIFT

Domain	Legacy Paradigm	MANGENESIS 2.0 Next-Gen Architecture
Geospatial Rendering	Static 2D Leaflet maps / raster tiles	WebGPU 4D volumetric terrain + PBR strata
Subsurface Modeling	2D drill hole log spreadsheets	Sparse 3D-CNN + Kriging Voxel Block Models
Fleet Telematics	End-of-shift operator tally sheets	Kafka streaming CAN-bus telemetry @ 10Hz
Shortfall Prediction	Reactive manual scheduling	LightGBM + InSAR & NASA GPM auto-calibration

2. End-to-End System Topology

1. Ingestion Tier

ISRO RISAT / Sentinel-2 / NASA GPM / CAT 785D IoT / Seismographs

2. Streaming & Event Bus

Apache Kafka / Redpanda Cluster + NATS JetStream (50k msgs/sec)

3. Data Lakehouse & GIS

PostgreSQL 17 (PostGIS 3.5) + TimescaleDB + MinIO GeoTIFF S3

4. AI & Analytical Microservices

Ray Cluster + 3D Kriging Voxel Engine + LightGBM Yield Predictor

5. Executive Frontend Tier

Next.js 15.2 (React 19) + WebGPU / Three.js r186 + Deck.gl GIS Engine

3. Next-Generation Frontend Architecture

The client-side architecture is re-engineered around React 19, Next.js 15.2 App Router, and a dedicated WebGPU/Three.js spatial pipeline.

3.1 WebGPU / Three.js 4D Spatial Viewport

Stack: [Three.js r186+](#) / [WebGPU](#) / [React Three Fiber](#) / [Drei](#)

- Hardware-accelerated procedural open-pit quarry with terraced bench geometry & normal bump mapping.
- Volumetric Subsurface Core Slicer: Real-time depth clipping planes down to -250m depth without CPU stall.
- Ore Voxel Block Model: GPU instance rendering of 250,000+ mineral grade blocks at consistent 60 FPS.
- Dynamic PBR lighting rig: Physical directional sun, atmospheric fog, and specular ore glint shaders.

3.2 Multi-Scale Geospatial GIS Engine

Stack: [Deck.gl v9](#) + [MapLibre GL v4](#) + [CesiumJS 3D Tiles](#)

- Massive vector overlay rendering: Displays 50,000+ drill collar locations, survey stations, and lease boundaries.
- InSAR Deformation Heatmap Layer: Millimeter-scale surface displacement contour visualizer.
- Continuous LOD (Level of Detail) tile streaming directly from Cloud-Optimized GeoTIFF (COG) endpoints.

3.3 State Management & Sub-15ms Telemetry Streaming

Stack: [Zustand v5](#) + [TanStack Query v5](#) + [WebSockets](#) / [WebTransport](#)

- Zustand for transient 3D camera matrices, layer visibility filters, and active zone inspection state.
- TanStack Query v5 for optimistic updates, background refetching, and offline cache warming.
- Binary Protobuf decoding over WebSockets for zero-lag CAT 785D haul truck positioning in 3D.

3.4 De-Congested Executive UI & Cyber-Luxury Design System

Stack: [Tailwind CSS v4](#) + [Radix UI](#) + [Framer Motion](#) ([OriginKit](#) / [Lightwind Style](#))

- Uncluttered, expansive layouts with generous breathing room (max-w-[1700px], 24px+ card padding).
- 3D Perspective Tilt Cards: Interactive cursor-following spotlight shaders and subtle metallic bevels.
- Minimalist floating frosted-glass pill controls (backdrop-blur-xl) over the 3D canvas instead of bulky black boxes.

4. Next-Generation Backend & Distributed Systems

The backend infrastructure leverages a modular microservices topology engineered for high-throughput spatial and telemetry ingestion.

Telemetry Ingestion Gateway

[Rust / Actix-Web]

Ingests CAN-bus J1939 logs from CAT 785D trucks, excavator hydraulic sensors, and seismographs @ 50k events/sec. Emits to Kafka.

Spatial Data Engine

[Go / Fiber + GDAL]

Processes satellite imagery (Sentinel-2, ISRO RISAT-1A), generates Cloud-Optimized GeoTIFFs (COGs), and serves PostGIS spatial queries.

Geostatistical AI Service

[Python 3.12 / FastAPI + Ray]

Executes Ordinary Kriging and 3D Sparse-CNN voxel interpolation for UNFC 111 reserve estimation and grade isosurfaces.

Predictive Production Engine

[Python 3.12 / LightGBM]

Predicts 7-day and 14-day production deficits calibrated continuously against radar precipitation, blast cycles, and haul delays.

Core Operations GraphQL API

[TypeScript / Node.js 22 (NestJS)]

Federated GraphQL & REST gateway providing RBAC-governed domain models, shift rosters, audit trails, and reporting.

4.2 Polyglot Persistence & Data Topology

- **PostgreSQL 17 + PostGIS 3.5:** Canonical Spatial & Relational Store — Mine leases, drill collars, assay logs, user entitlements, asset registry
- **TimescaleDB / ClickHouse:** High-Velocity Time-Series Lake — 100M+ sensor events, truck GPS tracks, vibration spectra, seismic monitors
- **MinIO / Cloud S3:** Geospatial Object Storage — Cloud-Optimized GeoTIFFs (COG), 3D Tiles Next, satellite multi-spectral bands
- **Redis 7.4 Cluster:** In-Memory State & Pub/Sub — Real-time truck coordinates, active alerts, distributed rate limiting

5. AI, Geostatistical & Geotechnical Pipeline

5.1 3D Subsurface Volumetric Grade Modeling

- Combines Variogram Modeling & 3D Ordinary Kriging with deep Sparse Convolutional Networks (Minkowski Engine).
- Interpolates continuous Mn grade, Fe impurities, and SiO₂ concentrations across 5m x 5m x 5m selective mining units (SMUs).
- Assigns JORC / UNFC 111, 121, and 122 confidence tiers based on geostatistical kriging variance.

5.2 Production Shortfall Forecasting (LightGBM + TFT)

- Ensemble model combining LightGBM gradient boosting with Temporal Fusion Transformers (TFT).
- Exogenous features: NASA GPM radar rainfall (mm/hr), excavator hydraulic pressure drops, haul road slickness indices.
- Generates probabilistic shortfall risk distributions (P10, P50, P90) up to 14 days ahead of processing plant feed.

5.3 Geotechnical Hazard & Slope Stability AI

- Persistent Scatterer InSAR (PSI) interferometry processing detects pit wall slope subsidence with millimeter precision.
- Continuous Factor of Safety (FoS) computation based on bishop simplified slice method integrated with groundwater piezometers.
- Automated alert dispatch to DGMS compliance dashboards when bench displacement rate exceeds 2.5 mm/day.

5.4 Optimal Ore Blending & Dispatch Optimization

- Mixed-Integer Linear Programming (MILP) solver optimizing daily pit extraction targets.
- Maintains uniform 44-46% Mn feed to the ferro-alloy plant while minimizing haul truck transit fuel by 8.4%.

6. Sovereign Security, Governance & Implementation Roadmap

6.1 Security & Sovereign Compliance Framework

- Zero-Trust Architecture: Mutual TLS (mTLS) across all inter-service communication via Istio Service Mesh.
- Sovereign Data Governance: All strategic mineral reserve data encrypted at rest (AES-256) and held in domestic Indian GovCloud.
- Indian Mines Act 1952 & DGMS Guidelines: Automated shift logging, statutory accident registers, and gas telemetry audits.
- UNFC 1997 / CRIRSCO Compliance: International standard geostatistical reserve reporting with audit logs.

6.2 Phased Engineering Execution Roadmap

Phase 1: Foundation & Spatial Core (Weeks 1 - 4)

Deploy PostgreSQL/PostGIS + TimescaleDB cluster; implement WebGPU 3D terrain viewer & drill-hole core visualizer.

' Deliverable: Interactive 3D Digital Twin with real geological PBR lithology & drill assay insp

Phase 2: Telematics & InSAR Pipeline (Weeks 5 - 8)

Deploy Rust/Go ingestion gateway, Kafka broker, and ISRO RISAT-1A / Sentinel-2 automated COG raster pipelines.

' Deliverable: Real-time 10Hz truck telemetry tracking in 3D + satellite multi-spectral alteration

Phase 3: AI Volumetric Kriging & Yield Prediction (Weeks 9 - 12)

Implement Ray-backed 3D Kriging voxel model, LightGBM 7-day shortfall predictor, and DGMS hazard alerts.

' Deliverable: Autonomous operational recommendations & UNFC 111 reserve estimation with 9

Phase 4: Multi-Mine Scaling & Hardening (Weeks 13 - 16)

Roll out across all 5 MOIL mines (Gumgaon, Dongri Buzurg, Mansar, Ukwa, Balaghat) with offline PWA field sync.

' Deliverable: Enterprise production sign-off, DGMS security audit clearance, and executive tr